



LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written examination

The use of electronic devices and additional material (books, notes, etc.) is not permitted during the exam. You may freely utilise all results proven in the lecture. Justify clearly your answers and please write your anonymous code at the top of each sheet.

- (a) Give an example of a group G and a G -set X such that X has exactly one G -orbit of cardinality p for each prime number p . (2 pts.)
(b) Let p be a prime number and G a group of order p^2 . Prove that G is abelian. (3 pts.)
- Determine all the Sylow subgroups of the symmetric group $\mathfrak{S}_3$. (5 pts.)
- Let R be a ring and suppose given a commutative square of R -module homomorphisms (indicated with solid arrows below) (5 pts.)

$$\begin{array}{ccccc} M & \xrightarrow{f} & N & \xrightarrow{p} & \text{coker } f \\ \downarrow u & & \downarrow v & & \downarrow \alpha \\ M' & \xrightarrow{g} & N' & \xrightarrow{q} & \text{coker } g \end{array} \quad v \circ f = g \circ u$$

where p and q are the canonical maps to the respective cokernels and u and v are isomorphisms of R -modules.

Prove that there exists a unique R -module homomorphism $\alpha: \text{coker } f \rightarrow \text{coker } g$ rendering the above diagram commutative (indicated with a dashed arrow above) and, moreover, that α is an isomorphism of R -modules.

- Let $\mathbf{k}$ be a field and $R := \text{Mat}_{2 \times 2}(\mathbf{k})$ the ring of 2×2 matrices with entries in $\mathbf{k}$. Fix $\lambda \in \mathbf{k}$ and consider the following subsets of the ring R : (5 pts.)

$$P := \left\{ \begin{pmatrix} a & -\lambda a \\ c & -\lambda c \end{pmatrix} \in R \mid a, c \in \mathbf{k} \right\}, \quad Q := \left\{ \begin{pmatrix} 0 & b \\ 0 & d \end{pmatrix} \in R \mid b, d \in \mathbf{k} \right\}.$$

Prove that P and Q are projective R -modules.

- (a) Let R be a ring and $M = S_1 \oplus S_2 \oplus \dots \oplus S_n$ a semisimple R -module. Prove or give a counter-example to the following statement: (2 pts.)

The semisimple R -module M admits only finitely many R -submodules.

- (b) Consider the ring $\mathbb{R}[X]$ of univariate polynomials with coefficients in the field $\mathbb{R}$ of real numbers. Prove that the $\mathbb{R}[X]$ -module S determined by the pair $(\mathbb{R}^2, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix})$ is simple. (3 pts.)

- Determine the elementary divisors and the invariant factors of the abelian group (5 pts.)

$$(\mathbb{Z}/30\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}/10\mathbb{Z}.$$